

Machine Vision & Intelligence Lab — NIT Jamshedpur

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Multilingual Abuse and Slang Detection

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Student Declaration

We hereby declare that the work presented in the report entitled ”**Multilingual Abuse and Slang Detection**” submitted by us for the partial fulfilment of the requirements for the degree of *Bachelor of Technology in Computer Science and Engineering* at NIT Jamshedpur, is an authentic record of my work carried out under the guidance of **Dr Koushlendra Kumar Singh**. Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Aditya Raj, Asim Kumar Mahata, Apoorva Singh, Mihir Kumar, & Abhishek Kumar
Place & Date: April 25, 2026

Certificate

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Dr Koushlendra Kumar Singh
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Place & Date: April 25, 2026

Abstract

In recent times the number of people going online is increasing day by day. People spend too much of their life behind screens. This leads to people encountering Toxic people which affects their mental health specially kids. To Fix this issue Social media sites take action to regulate this behavior but this problem is mostly target towards people who know english . So toxic comments which are code-mix , regional language or slang pass their detection system making it ineffective in these scenarios.

So In our project, we propose a Multilingual Abuse and slang detection model using XLM-RoBERTa-large backbone , which is trained on 100+ languages specially Indian languages. Since in India there are so many regional languages as well as their transliterations it's very common for people to use Code-switched or mixed languages. in this project we fined tuned this model and trained it on 12 different language variants which are majorly used in India making it a suitable choice to moderate this portion of the problem.

Our proposed Model Not only detect the toxicity of an comment, it also provide a language ID which helps in further context about the toxicity of the comment. As well as providing Attention Weights and Lime Token visualization to detect the exact word that triggered the toxicity flag. Thus Making it one of the most reliable ways to moderate this kind of toxic behaviors on online platforms.

Our model perform well in terms of accuracy and f1 score on not only individual languages but also Overall accuracy and f1 score.

Our implementation includes data collection, data pre-processing, model training, converting into onnx model for fast query , a REST API to get fast result, and a frontend to demonstrate our model.

Keywords: Multilingual NLP, XLM-Roberta, Code-switching, Hate Speech Detection, Language Identification, Real-time Moderation, Attention Mechanisms.

Acknowledgment

We would like to express our sincere gratitude to our guide, **Dr. Koushlendra Kumar Singh**, for their constant guidance and his suggestions which plays a big part during the making of this project. Dr. Koushlendra Kumar Singh's expertise in Digital Image Processing and Natural Language Processing was a great help for us in deciding our strategy and dealing with the issue faced during our work of multilingual toxicity detection.

We would also want to thanks to the **Machine Vision & Intelligence Lab(MVI Lab) and the Department of Computer Science and Engineering, NIT Jamshedpur** for giving us the necessary facilities for this project.

And at last, we would also like to acknowledge the contributions of the teammates involved in making this project a success.

Work Distribution

The responsibilities for the project "Multilingual Abuse and Slang Detection" were distributed among the team members as follows:

- **Aditya Raj:**Project coordination, Frontend and backend development, REST API integration, and deployment of the system on HuggingFace, model testing.
- **Asim Kumar Mahata:** Data collection across major Indian languages and implementation of the training pipeline.
- **Apoorva Singh:** Literature review and researching existing approaches for multilingual hate speech detection,report writing.
- **Mihir Kumar:** Planning the model architecture and preprocessing pipeline design.
- **Abhishek Kumar:** Frontend and backend development, ONNX export for fast inference, and attention map generation,model testing.

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Chapter 1

Introduction

1.1 Background

Online hate speech is quite prevalent in our today's internet/social media heavy world. Spreading hate through online tweets, comments, videos, reels etc is quite an issue. Hate speech, cyber bullying, and toxic language detection manually is practically impossible with the amount of digital footprints being produced everyday. We were asked to propose an automated solution for this issue. So, our project focuses on building an automated system for detecting hate speech, cyber bullying, and hate language usage.

But today's social media platforms like x.com(twitter) and instagram comments don't have people of only one language/background interacting. They are of different backgrounds, use different languages, use combination of languages like hindi + english, bangali + english, tamil + hindi, tamil + english etc etc. They often use abbreviations and informal spellings to avoid auto detection, use hate emojis, use hate gifs etc etc. Traditional detection models are not capable of working with these mix of languages. They are mostly built for english, often mistaking the contexts when the hate is occurring in other languages. This also influenced our project's direction.

The main goal of our project is to identify hate/rude online text content by not losing context of the language in which the hate is occurring. It is happening by identifying the offensive text content across multiple languages and across multiple combinations of languages.

Chapter 2

Literature Review

2.1 SemEval-2019 Task 5: Multilingual Detection of Hate Speech Against Immigrants and Women in Twitter[1]

SemEval, it is an international level natural language processing competitions and research workshops. It focuses on moving humanity forward by improving the current state of the art in semantic evaluation. SemEval literally means semantic evaluation. This SemEval 2019 was one such competition.

Task 5: multilingual hate speech detection. That was the task number 5 of that year's competition/workshop.

Dataset: They were provided with datasets in two different languages. English and Spanish.

- **Subtask a:** binary classification of if the speech was hate speech or not
- **Subtask b:** fine grain classification to identify if the hateful speech was against a community/group of people or against just a single person.

Results concluded in the paper:

- Subtask b was a lot harder to get accurate results of than of subtask a.
- Spanish trained models performed better.

2.2 Detection of Hate Speech Text in Hindi-English Code mixed Data[2]

This paper is a lot more relevant to our project as it actually focuses on the Indian based languages. This paper talked about the difficulty of predicting hate speech in a code mixed language like Hinglish, Tanglish etc. This paper talked about Hinglish basically. It's the most prevalent on Indian social medias.

Dataset: this paper used a dataset of 10,000 samples of speech taken from various sources like x.com etc. it was in hinglish code mixed version. **feature extraction:** They tried/represented 3 different methods to extract features from the dataset. those were

- fastText
- word2vec
- doc2vec

Results:

- best performing one was fastText iwth 85.81% , they used SVM with rbf kernel.
- word2vec had highest achieved accuracy of 75.11% , it also used SVM with rbf kernel.
- doc2vec highest achieved accuracy of 64.15% classifier used was random forest.

They concluded that character-level features used by fastText are superior for code mixed text . It can handle non standard spellings very well.

Paper Title	Publisher	Authors	Year	Key Result
SemEval-2019 Task 5: Multilingual Detection of Hate Speech Against Immigrants and Women in Twitter	ACL Anthology (SemEval-2019)	Basile V., Bosco C., Fersini E., Nozza D., Patti V., Rangel F., Rosso P., Sanguinetti M.	2019	fine grain classification into hate type
Detection of Hate Speech Text in Hindi-English Code-mixed Data	Procedia Computer Science (Elsevier)	Sreelakshmi K., Premjith B., Soman K.P.	2020	code mix language hate speech detection

Table 2.1: Summary of reviewed literature on hate speech detection

Chapter 3

Methodology

Step 1: Data Collection / Dataset Preparation:

Data Sources:

Data from different sources are combined to make a single dataset, containing different languages. - English, Hindi, Hinglish, Bengali, Banglish, Marathi, Kannada, kanglish, Tamil, tanglish, Malayalam, maglish.

Data uniformity:

all the data gets a uniform format before going for preprocessing, the final single dataset is a tsv containing 3 columns, which include text, offensive (0 or 1), language. The data is then shuffled to do the proper training without any bias of any language.

Balancing:

In order to make model robust and not get skewed toward a specific language, or class, the data is balanced in the dataset. The difference between the number of rows of two language is tried to keep it minimum, and the no. of offensive and non-offensive classes are almost 50-50 in all the language, so that it do not show biasness toward a class.

Stratification:

During the sampling process, stratified sampling preserves the original distribution of the data.

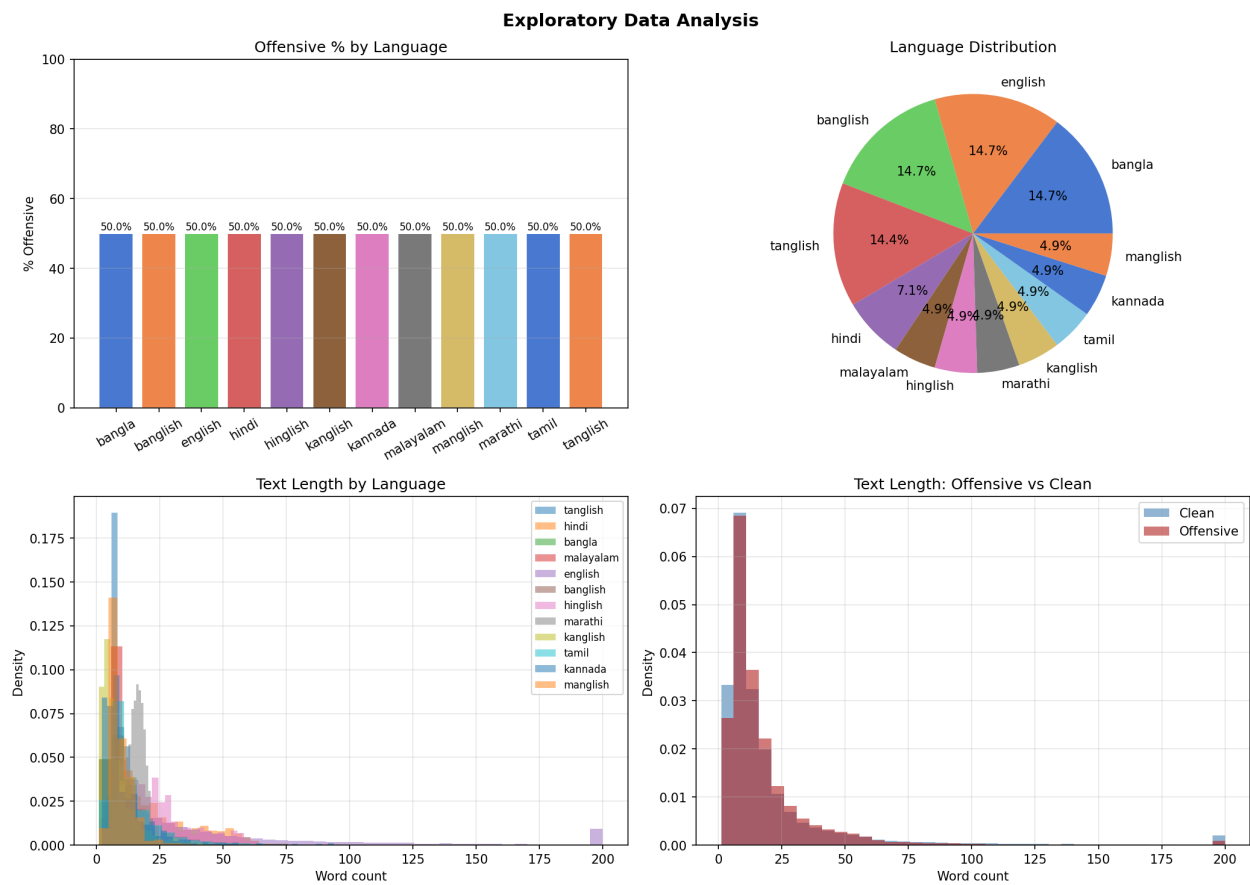


Figure 3.1: Language Distribution in dataset

Step 2: Data Preprocessing and Feature Engineering

The Preprocessing stage focuses on transforming raw, noisy multilingual data taken directly from sources like x.com and datasets that were mentioned in the previous step into a clean, uniform format suitable for the XLM-RoBERTa-large model.

- **Noise Removal:** Stripped URLs (HTTP/WWW), HTML tags, and redundant metadata.
- **Whitespace Control:** removed multiple empty lines into one single line break and empty spaces into single space
- **Case Folding:** all text was converted into lowercase letters so that all words are emphasized on equally
- **Emoji To text:** Converted visual icons into text to preserve context necessary for the hate identification
- **Abbreviation Expansion:** Expanded common English shortcuts (like "wtf", "idk") into their full forms
- **Multilingual Slang Mapping:** Used specialized dictionaries to map regional slang (Hinglish, Banglish, Tanglish, etc.) to standardized terms, preventing loss of cultural context.

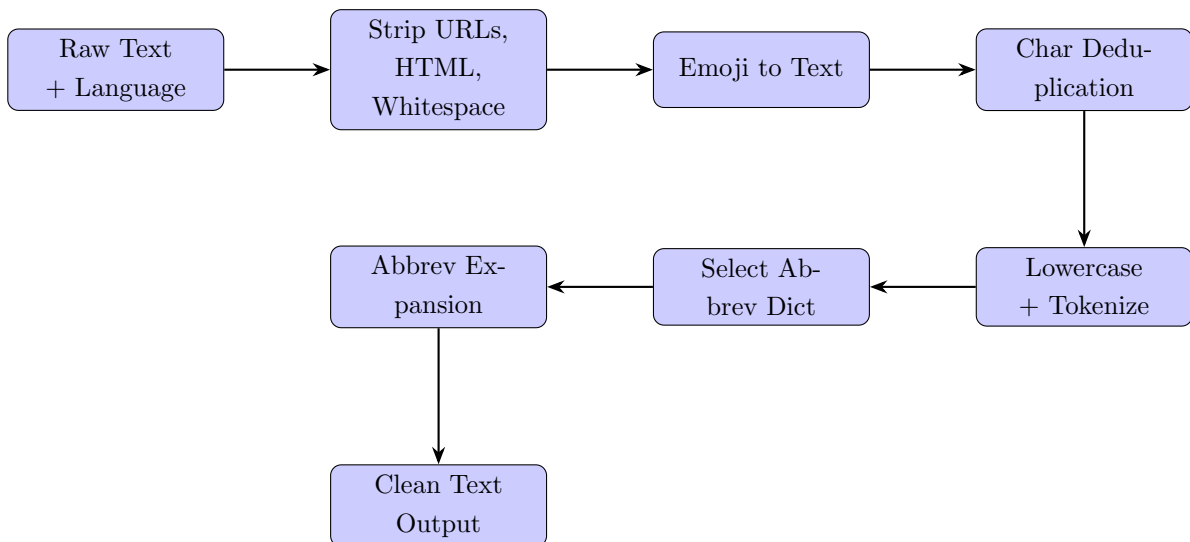


Figure 3.2: Preprocessing pipeline.

Step 3: Model Design/Algorithm Selection

The multitask model implementation focuses on predicting both offensive content detection and language identification across multilingual datasets.

Model Architecture

1. **Single shared backbone:** The main architecture is XLM-RoBERTa-large, a transformer model trained on 100+ Indian languages, it generates 1024 dimension CLS vector for the input text.
2. **Offensive head detection:** The output from the backbone is put into a linear classification followed by sigmoid activation function for predicting whether the text is offensive or not. Binary cross entropy loss with logits is being used for optimizing the head.
3. **Language detection head:** At the same time, the embedding passed to another linear head with softmax activation function, for detecting the language of the text. It is optimized using cross entropy loss.

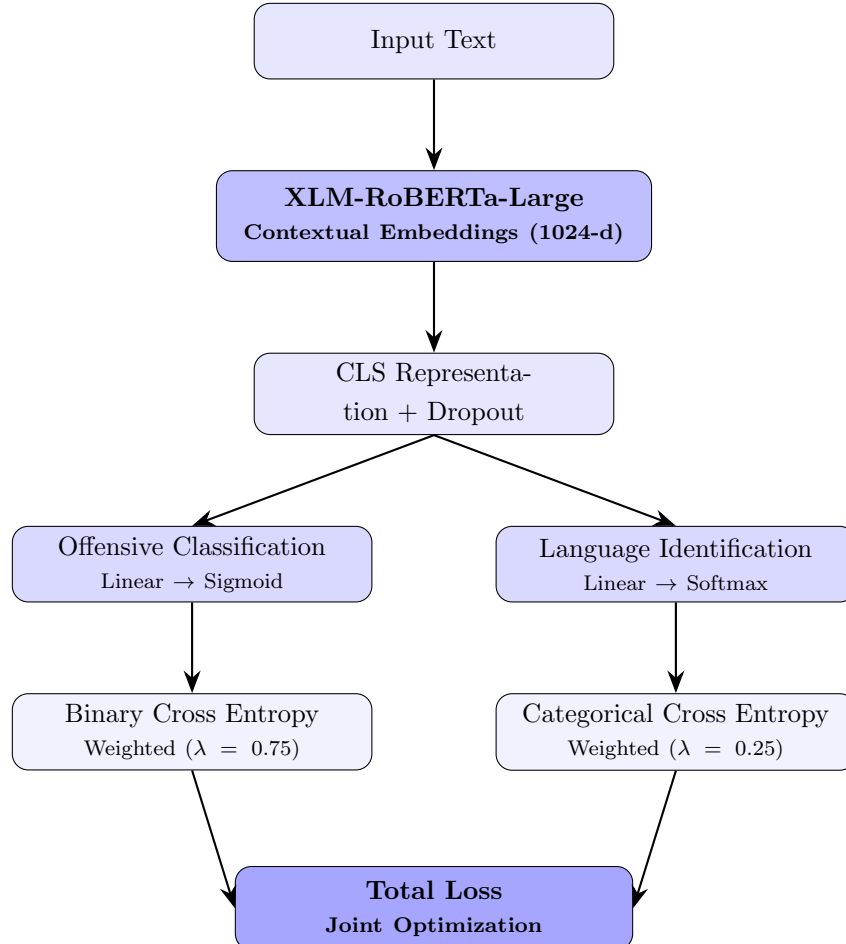


Figure 3.3: Multi-task architecture Using XLM-RoBERTa-large Backbone.

Step 4: Training and Optimization

1. **Loss:** A multi-class loss function is used which is calculated as 75% of offensive loss + 25% of language loss, so that offensive detection head got priority over language detection head.
2. **Optimization:** AdamW optimizer is used with the learning rate of 1e-5. The weight decay is 0.01 and 10% of warmup ratio for dynamically changing the learning rate.
3. **Memory Efficiency:** Since XLM-RoBERTa-large model is big, to handle that base batch size of 8 is used, and gradient accumulation steps of 4 is used, which results in effective batch size of 32.
4. **Validation and Early stopping:** The data is divided into 90% training, 10% validation, 10% testing, stratified by both language as well as labels. Early stopping with patience of 5 epochs are used to stop training if model is not showing any improvement.
5. **Best Model Selection:** The training process automatically identifies and saves the "Best" model based on the highest achieved Validation Macro F1 score.

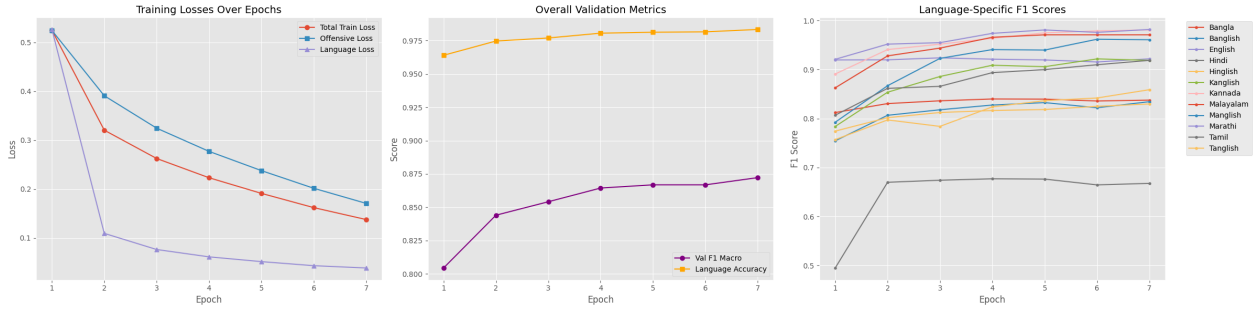


Figure 3.4: Training Logs

Step 5: Evaluation and Testing

1. **Dynamic thresholding:** Instead of keeping decision boundary as something like 0.5 for offensive categorization, the model threshold is dynamically tuned. It is swept here and there in order to find where the value which maximised the macro F1 score in the validation set.
2. **Metrics and evaluation:** The model performance is being evaluated by various metrics like accuracy, macro/micro/weighted F1, ROC-AUC precision and Matthews correlation coefficient.
3. **Per language breakdown:** After calculating the evaluation for the overall model, it is then calculated for each of the language also so that all of them work well, and there is equality among them.

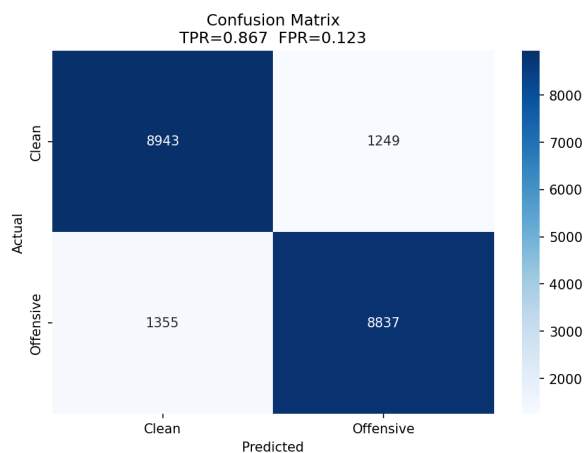


Figure 3.5: Confusion Matrix (Toxicity)

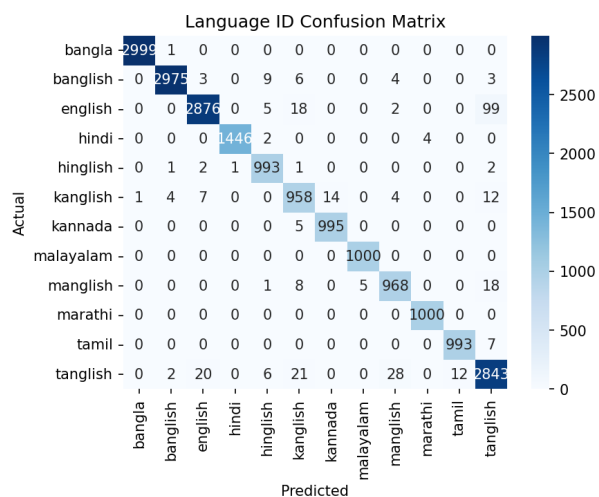


Figure 3.6: Confusion Matrix (Language ID)

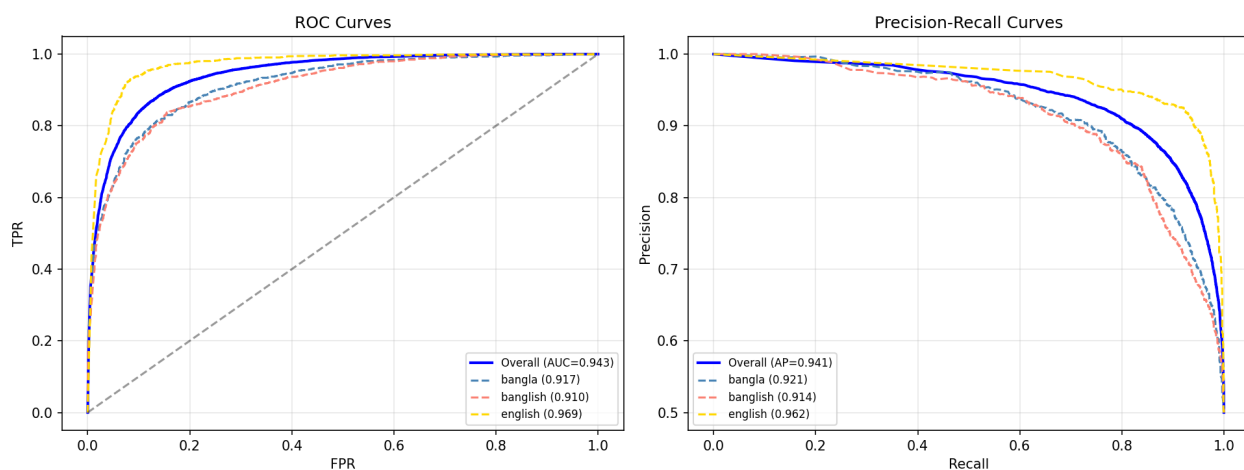


Figure 3.7: Roc and Precision,Recall curves

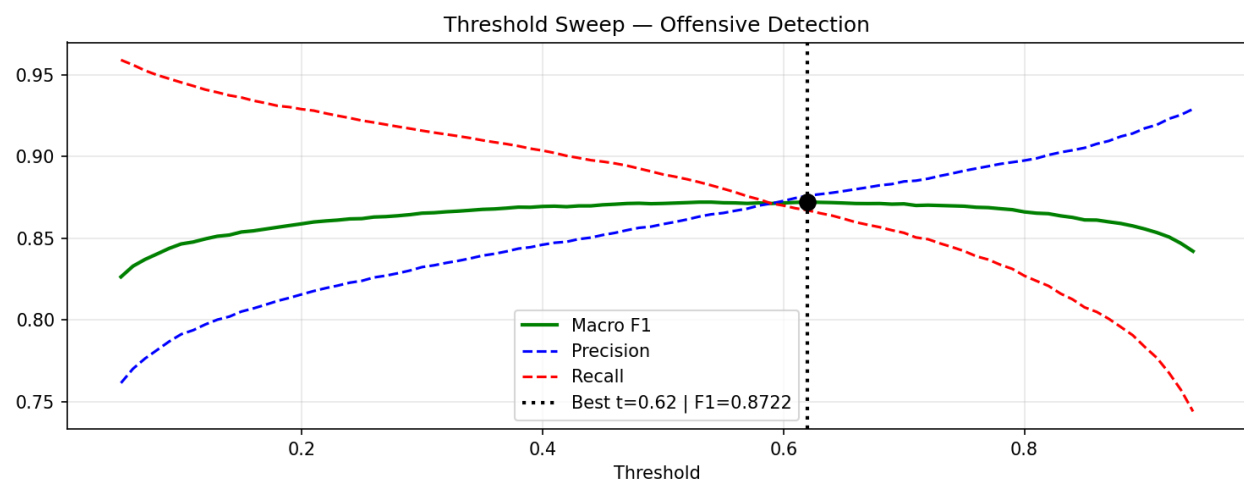


Figure 3.8: Threshold Tuning

Step 6: Model Deployment

- A web based interface was developed using Streamlit to provide an interactive platform for testing the model in real time.
- This interface allows users to input multilingual text and receive immediate feedback, including the offensive verdict, the detected language, and model confidence scores.

Endpoint	Description
/predict	Returns the toxic flag and language ID
/pipeline	Return the various steps of process
/explain_lime	LIME token-level attribution
/explain_attention	Raw attention matrix from last layer
/explain_word_attention	CLS-row attention merged at word level

Table 3.1: FastAPI backend endpoints.

- The model is exported to ONNX format, for faster inference, with PyTorch as fallback.
- LIME and attention weight visualizations are added to show which words triggered the toxic flag in the sentence.

→ **The frontend interface is accessible at:** <https://multi-lang-abuse-detection.streamlit.app/>

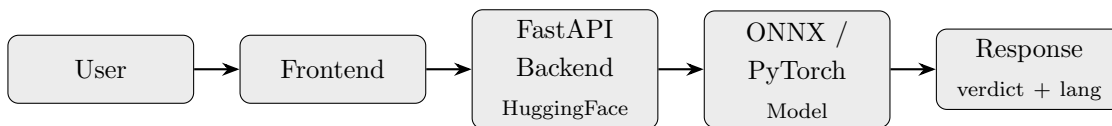


Figure 3.9: Deployment pipeline.

Chapter 4

Results

Overall Performance

Table 4.1 is the metrics of offensive head of the overall model. The model achieves accuracy of 87.23 % and macro f1 score of 0.8722.

Metric	Score
Accuracy	0.8723
Macro F1	0.8722
Micro F1	0.8723
Weighted F1	0.8722
ROC-AUC	0.9433
Avg Precision	0.9405
MCC	0.7445

Table 4.1: Overall offensive detection metrics.

Language Specific Performance

Table 4.2 breaks the model performance for all the languages. The model is performing well in most of the native languages as well as in the code-switched languages.

The weakest performance is in hindi language, this area need to get some upgrade with more data collection and targeted fine tuning.

Overall, the model is generalizing well across all the languages without any biasness toward a specific language.

Language	Samples	Macro F1	Micro F1	ROC-AUC
Kannada	1000	0.9810	0.9810	0.9946
Marathi	1000	0.9820	0.9820	0.9955
Malayalam	1000	0.9710	0.9710	0.9964
Manglish	1000	0.9610	0.9610	0.9898
English	3000	0.9220	0.9220	0.9695
Kanglish	1000	0.9189	0.9190	0.9618
Tamil	1000	0.9189	0.9190	0.9595
Hinglish	1000	0.8600	0.8600	0.9205
Bangla	3000	0.8377	0.8377	0.9170
Banglish	3000	0.8345	0.8347	0.9100
Tanglish	2932	0.8301	0.8302	0.8881
Hindi	1452	0.6670	0.6680	0.7483
Overall	21384	0.8722	0.8723	0.9433

Table 4.2: Per-language offensive detection metrics (sorted by Macro F1).

Chapter 5

Conclusion

The project is multilingual abuse and slang detecting system, which is capable of detecting abusive content across many languages, which are used in Indian context, along with their romanized and code-switched forms such as Hinglish, Banglish, Tanglish, etc. The system is build on xlm-roberta-large as backbone, and this single model is acting as backbone with two different heads, one is for abusive detection and other is for language detection. training strategies such as weighted loss, AdamW optimization, gradient accumulation and early stopping is use for efficient training.

The model got an overall Macro F1 score of 0.8722 and ROC-AUC of 0.9433 on the test set. Native languages such as Kannada, Malyalam, Marathi shows good performance while Hindi is still challenging one.

The system is then deployed as a REST API using FastAPI, with ONNX export for the inference. Explainability features with LIME attribution and attention weight visulaisation is also used for getting which word is trigerring the toxicity.

At last, the project is a single multilingual model that is used to moderate offensive content over diverse user base, which most of the current existing moderation system fails to cover.

Chapter 6

References

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